A Comparative Study of Self-Supervised and Spectrogram-Based Models for Cross-Corpus Speech Dysfluency Detection

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Abstract. Stuttering is a speech disorder that manifests as involuntary repetitions, prolongations, and blocks, impacting communication and quality of life, and it affects roughly 1% of the adult population and up to 5% of children [16]. Automated detection of dysfluency events is challenging due to the subtle acoustic signatures and strong class imbalance across dysfluency types (prolongation, block, sound repetition, word repetition, interjection). This paper presents an end-to-end speech dysfluency classification system and compares seven experimental configurations spanning two representation families: self-supervised waveform-based models built on Wav2Vec 2.0 (five independent binary classifiers and a shared-backbone multitask variant) and compact mel-spectrogram CNN models (pooled, single-head, LSTM-based, and transformer-based). All models are evaluated on a merged corpus of 37,087 clips drawn from SEP-28K, UCLASS, and Project Boli (36,674 usable after removing empty audio) under two protocols: an in-distribution clip-level held-out evaluation and a speaker-independent cross-corpus evaluation on Project Boli (unseen speakers and recording conditions; the ingested Boli clips are English-only, so this is a cross-corpus but not a cross-lingual evaluation). Under the clip-level protocol, the five-binary Wav2Vec 2.0 configuration achieves the highest measured macro F1 (0.5704 at validation-tuned thresholds), while the shared-backbone multitask variant follows closely (0.5557) using $\approx 4.9\times$ fewer parameters than five separate classifiers; however, this protocol shares speakers between training and test, so these results are not speaker-independent. On the 53-clip speaker-independent Boli set, no configuration exceeds a trivial all-positive baseline (macro F1 $= 0.55$): apparent cross-corpus F1 differences are dominated by class base rates and by threshold/operating-point drift on the distribution-shifted set rather than by ranking quality. Under the threshold-free metrics the ordering is reversed, with the two task-adapted Wav2Vec2 configurations ranking Boli clips best (macro AUROC ≈ 0.62) while the CNN configurations sit around chance (≈ 0.47–0.52). These results demonstrate that tuned-F1 comparisons on a small imbalanced external corpus do not reliably measure cross-corpus generalization; we therefore treat ranking-based metrics and per-class analysis as the primary evidence, discuss the trade-offs between representation capacity and generalization, and frame the observed generalization gap as an open hypothesis. We report per-class precision, recall, F1, AUROC, and AUPRC across all five dysfluency types. All results are single-seed point estimates and should be interpreted accordingly.

Keywords: Speech dysfluency detection · stuttering classification · Wav2Vec 2.0 · multitask learning · cross-corpus evaluation · speaker-independent evaluation.

1 Introduction

Stuttering affects approximately 1% of the global adult population and up to 5% of children [16]. Characterized by involuntary repetitions of sounds, syllables, or words (sound repetitions and word repetitions), abnormal prolongations of speech sounds (prolongations), and involuntary pauses or stops in speech flow (blocks), it impacts the psychosocial well-being of affected individuals [16]. Traditional assessment by speech-language pathologists (SLPs) relies on subjective auditory perception and visual inspection of waveforms, a process that is time-consuming, prone to inter-rater variability, and unscalable for large populations; this has motivated a large body of work on automatic stuttering identification [14].

Recent advances in self-supervised speech representation learning, particularly Wav2Vec 2.0 [2], have achieved strong performance on downstream speech classification tasks by learning contextualized representations from raw audio. Applying such models to stuttering detection presents challenges: the five dysfluency types have different acoustic signatures, and severe class imbalance exists (most speech segments are fluent) [12].

This paper makes the following contributions:

- We present a modular system for speech dysfluency classification, supporting five dysfluency types: prolongation, block, sound repetition, word repetition, and interjection.
- We conduct a systematic comparative study of seven experimental configurations across two representation families—self-supervised waveform models (Wav2Vec 2.0) and mel-spectrogram CNN models—on a merged multi-corpus dataset.
- We evaluate under two complementary protocols: a clip-level in-distribution held-out split and a speaker-independent cross-corpus held-out set (Project Boli), and we report both macro and per-class metrics including AUPRC.
- We analyze per-class performance across all five dysfluency types, showing that interjection achieves the highest F1 among the evaluated classes while block is the most challenging. We further demonstrate that tuned-F1 results on the small external Boli set are dominated by class base rates and operating-point drift—no configuration beats a trivial all-positive baseline there—so cross-corpus claims must be evaluated with threshold-free metrics, and we articulate the open hypotheses that surround the observed in-distribution/cross-corpus generalization gap.

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 describes the system architecture, Section 4 details the experimental setup, Section 5 presents results and discussion, and Section 6 concludes.

2 Related Work

2.1 Traditional Approaches

Early stutter detection systems relied on handcrafted acoustic features, including jitter, shimmer, spectral flux, and formant trajectories, combined with classical classifiers such as Support Vector Machines (SVMs), Hidden Markov Models (HMMs), and

Gaussian Mixture Models (GMMs) [14]. These methods achieved reasonable accuracy on controlled datasets but generalized poorly across speakers, accents, and recording conditions due to the limited representational capacity of handcrafted features.

2.2 Deep Learning Methods

The introduction of end-to-end deep learning approaches removed the need for manual feature engineering. Convolutional Neural Networks (CNNs) operating on mel-spectrograms, combined with recurrent layers, showed early promise for multi-type dysfluency detection [11]. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and time-delay neural networks were applied to capture temporal dependencies in speech signals [11, 13]. More recently, transformer-based architectures, particularly Wav2Vec 2.0 [2], have achieved strong results on speech processing tasks by learning contextualized representations from raw audio through self-supervised pre-training on large unlabeled corpora.

2.3 Self-Supervised Speech Representation Models

Self-supervised pre-training has become the dominant paradigm for speech representation learning. Wav2Vec 2.0 [2] learns contextualized speech representations by contrasting quantized latent speech units and masking spans of the input waveform, and it generalizes across languages when pre-trained on multilingual data (XLSR-53 [7]). HuBERT [10] instead predicts masked hidden units obtained from an offline clustering step and has been shown to capture strong segmental and suprasegmental characteristics of speech. XLS-R [1] scales the Wav2Vec 2.0 objective to 128 languages and 436K hours of speech. WavLM [6] augments masked prediction with denoising objectives to improve robustness on full-stack speech-processing downstream tasks. A consistent property of these models is that their embeddings encode both linguistic content and paralinguistic attributes such as speaker identity and recording condition; the amount of such non-lexical information carried in the representation is a design trade-off that is central to our analysis of cross-corpus generalization.

2.4 Wav2Vec2-Based Stutter Detection

Bayerl et al. [4] applied Wav2Vec2 to multi-task stuttering detection on FluencyBank, achieving macro F1 scores in the range of 0.56–0.63 across five dysfluency types. On the standardized SEP-28k-E partition, published state-of-the-art results reach per-class F1 values that vary significantly by dysfluency type (e.g., interjection: 0.78, prolongation: 0.53, block: 0.30, as reported by Sušac et al. [15] for the leading baselines). These works established Wav2Vec2 as a strong backbone for stutter detection but focused on classification within a single corpus without systematic architectural comparison or cross-corpus evaluation.

2.5 Stuttering and Dysfluency Datasets

Several stuttering corpora have become available in recent years. SEP-28K was introduced by Lea et al. [12] from publicly available podcasts of people who stutter, with clip-level majority-vote labels for five dysfluency types. The same authors released a compatible re-annotation of the adults-who-stutter portion of FluencyBank, a corpus of transcribed monologue and dialogue speech [5]; together these two sources provide the 32,321 SEP-28K-format clips used here. UCLASS, a corpus of stuttered speech from therapy sessions [9], provides 4,712 additional clips in SEP-28K format. Project Boli [3] is a recent multilingual, multi-accent corpus with word-level dysfluency annotations in five Indian languages. These datasets differ in speakers, languages, recording conditions, and annotation granularity (clip-level versus word-level), which makes cross-corpus generalization difficult to assess and rarely tested.

2.6 Cross-Corpus and Speaker-Independent Evaluation

Prior work has largely evaluated on a single dataset (either SEP-28K or FluencyBank), making it difficult to assess cross-corpus generalization. A system trained on one corpus may overfit its speakers, accents, and recording conditions; evaluating on held-out speakers from the same corpus only partially reveals this when speakers overlap across splits. Reporting a speaker-independent protocol (no speaker in more than one split) and a cross-corpus protocol (an entirely separate corpus) is therefore important for realistic performance estimates. We follow this reasoning and evaluate all seven configurations both in-distribution (clip-level) and cross-corpus on Boli, whose speakers and recordings are fully disjoint from training.

3 System Architecture

The system comprises a classification pipeline that identifies which dysfluency types are present in speech audio. The pipeline builds on a common preprocessing module and exposes models through a unified model registry API.

3.1 Data Pipeline

The system integrates three publicly available stuttering datasets:

- *SEP-28K* [12]: 32,321 clips annotated for five dysfluency types (28,177 podcast clips plus a 4,144-clip FluencyBank re-annotation), with TSV-format labels.
- *UCLASS* [9]: 4,712 clips in SEP-28K format, sourced from stuttering therapy sessions.
- *Project Boli* [3]: 54 clips with word-level dysfluency annotations and timestamps, drawn from the corpus's English task recordings. Boli is a multilingual, multi-accent corpus whose full release carries word-level annotations in five Indian languages, but our ingestion path resolves the English task codes only, so the 54 clips used here are all English. The cross-corpus evaluation reported below is therefore a change of speakers, recording pipeline, and annotation convention—not a change of language.

Table 1. Source corpora and their integration into the merged corpus. The Boli clips used here are the English portion of a corpus that also contains Hindi, Marathi, Telugu, Bengali, and Assamese material; the non-English portions were not ingested.

Dataset	Clips	Language(s)	Annotation	Role
SEP-28K	32,321	English	clip-level TSV, majority vote	train/val/test
UCLASS	4,712	English	clip-level TSV	train/val/test
Boli	54	English	word-level intervals	test only

Table 1 summarizes their key properties. The three datasets are merged into a unified corpus of 37,087 clips with per-clip interval CSVs recording dysfluency events as `(start_sec, end_sec, type)` tuples. Clips with missing or header-only (empty) audio files are removed, leaving 36,674 usable clips. An 80/10/10 random split (seed 42) assigns 29,296 training clips, 3,662 validation clips, and 3,716 test clips. To guarantee an honest cross-corpus evaluation, all Project Boli clips are pinned to the test split and never appear in the training or validation splits. Audio is resampled to 16 kHz mono, DC-removed, peak-normalized, silence-trimmed, and padded/truncated to a maximum of 3 seconds.

We note one departure between this split and the one the reported models were actually trained and tuned on. The prepared 3,662-clip validation split is not consumed by the training or evaluation code; both read the training directory and derive their own held-out portion from it. Training and threshold selection therefore operate on a 20% stratified subsample of the 29,296 training clips (5,857 clips, seed 42), stratified by the first positive dysfluency class. The prepared validation directory is unused in every number reported here. We describe the operating point as *validation-tuned* throughout, but the reader should understand it as tuned on a held-out subsample of the training pool rather than on the nominal validation split.

Annotation harmonization. The three corpora use different annotation conventions, which we map onto a common set of five binary clip-level labels (prolongation, block, sound repetition, word repetition, interjection). SEP-28K labels are adopted as-is: a clip is positive for a dysfluency type if at least two annotators marked it (`SEP28K_MIN_VOTES` = 2). UCLASS provides one label set per clip in the same five classes; because each clip is a 3-s window centered on the stutter event, each present label is materialized as a weak interval of half-width 0.25 s centered at 1.5 s into the clip, i.e. spanning 1.25–1.75 s. Project Boli provides word-level tokens annotated with the labels *B* (block), *PR* (prolongation), *SR* (sound repetition), *WR* (word repetition), and *IN* (interjection); these are mapped through a fixed code table to the same five classes, and a clip is labeled positive for a class if any of its annotated words carries that class. Thus all downstream models consume identical five-dimensional binary targets regardless of the original corpus.

The 3-second window. SEP-28K clips are natively 3 s long, so a maximum window of 3 s is the natural budget for the classification task and matches the predominant corpus. For UCLASS, whose clips are 3-s windows centered on a stutter event, the window preserves the event of interest. Because classification is clip-level, an event whose acoustic extent crosses the window boundary does not alter the training target: the clip-level label only records presence or absence. This is not the case for temporal localization, which is out of scope here. Two caveats apply. First, aggressive silence trimming followed by zero-padding to 3 s can place leading or trailing speech at the window edges, discarding some prosodic context. Second, for blocks—which manifest as silent or tenuous pauses—the fixed window and the trimming may suppress the very pause that defines the event when it falls near a boundary. We return to this in the discussion of per-class results.

3.2 Classification Pipeline

Five Independent Binary Classifiers. Each dysfluency type is handled by a separate binary classifier built on `facebook/wav2vec2-base` (94,569,090 parameters per classifier, including a $\approx$0.2M-parameter classification head on top of the 94.37M-parameter backbone). The architecture follows the sequence-classification formulation of `Wav2Vec2ForSequenceClassification`: raw audio $\rightarrow$ Wav2Vec2 encoder $\rightarrow$ a learned projection of the frame-level hidden states (`Linear`(768$\rightarrow$256)) $\rightarrow$ mean pooling over time $\rightarrow$ a linear classifier (`Linear`(256$\rightarrow$2)) $\rightarrow$ softmax. Two details of the reference implementation are worth stating explicitly because they are easy to assume otherwise: the projection precedes the pooling rather than following it, and the head contains no dropout layer. No attention mask is passed by our pipeline, so the temporal reduction is an unmasked mean over frames; because all clips are zero-padded or truncated to a fixed 3 s, every clip presents the same number of frames. Each classifier is trained independently with focal loss ($\gamma = 2.0$) to handle class imbalance, the AdamW optimizer (lr $= 3 \times 10^{-5}$, weight decay $= 0.01$), a linear warmup (500 steps), and mixed-precision training on CUDA. Backbone freezing for the first 3 epochs prevents catastrophic forgetting of the pretrained representations; once the backbone is unfrozen it is trained with a reduced learning rate of 3×10^{-6} (one tenth of the head rate).

Shared-Backbone Multitask Classifier. A single Wav2Vec2 encoder feeds five parallel per-class heads, each a small MLP (`Linear`(768$\rightarrow$768) $\rightarrow$ Tanh $\rightarrow$ `Linear`(768$\rightarrow$2)) operating on a temporal mean pool of the encoder's hidden states. The total parameter count is 97.3M, only $\approx 2.8M$ more than a single independent classifier, since the backbone is shared. Training uses the summed focal loss across all five heads, allowing the shared backbone to learn representations useful for all dysfluency types. The same 3-epoch freezing schedule and post-unfreeze learning-rate reduction as arm01 are used.

CNN Spectrogram Models. The compact CNN models (arms04–07) share a common backbone but differ in their sequence aggregator. The input is a log-mel spectrogram of 128 Mel bands computed with a 2048-point FFT and a hop length of 512 on the 16 kHz audio. The encoder is four convolutional blocks with 32, 64, 128, and

128 channels respectively, each using a 3×3 kernel with padding 1, batch normalization, ReLU activation, a $(2, 1)$ max-pool (all but the last block), and 2-D dropout with rate 0.4. The encoder output is a $128 \times 16 \times 94$ tensor (channels $\times$ mel bins $\times$ frames), since the $(2, 1)$ pool halves the frequency axis three times and leaves the temporal axis at 94 frames; this is passed to one of three aggregators: (i) *pool*—global mean pooling over time and frequency followed by a `Linear`(128→128) projection; (ii) *LSTM*—a single-layer unidirectional LSTM with a hidden size of 128 operating over the frequency-pooled time axis, whose final hidden state is used; or (iii) *transformer*—a one-layer transformer encoder with a model dimension of 128, four attention heads, a feed-forward width of 512, and dropout 0.4, applied over the frequency-pooled time axis and followed by temporal mean pooling. Finally, five per-class two-layer MLP heads (`Linear`(128→128) → Tanh → `Linear`(128→2)) produce the logits for each dysfluency type. arm04 uses the pool aggregator (342,030 parameters), arm06 the LSTM aggregator (457,614), and arm07 the transformer aggregator (540,302). arm05 is the single-head variant of the pool model (274,950 parameters): the head block is built once per configured class rather than five times, so each arm05 model carries the same two-layer head (`Linear`(128→128) → Tanh → `Linear`(128→2)) but only a single instance of it, and one such model is trained per dysfluency type (five models in total), mirroring the five-binary Wav2Vec2 design at a much smaller scale.

The CNN arms are trained from scratch, so no backbone freezing or discriminative pre-training applies. Their loss differs from the Wav2Vec2 arms: instead of focal loss, they use binary cross-entropy on the positive logit with a per-class positive weight set to the negative-to-positive class ratio of the training pool, and the per-head losses are summed over the configured classes, mirroring the multitask Wav2Vec2 objective. The two families therefore differ in training objective as well as in representation, which we note as a confound when comparing them.

3.3 Model Registry

All classifiers are loaded through a unified registry module (`Classifier`) that decouples model loading from training. The registry reads checkpoint paths and per-class thresholds from a JSON configuration file, enabling model swapping without code changes. Audio preprocessing (resampling, cleaning, normalization, padding) is applied automatically within the API.

4 Experimental Setup

4.1 Model Configurations

We evaluate seven experimental configurations, summarized in Table 2. The first configuration consists of five independently trained binary Wav2Vec2 classifiers, one per dysfluency type; the remaining configurations are single models.

Table 2. Experimental configurations, aggregator types, and parameter counts.

Arm	Architecture	Type	Parameters
arm01	5× Wav2Vec2 binary classifiers	classifier	472,845,450
arm02	Wav2Vec2 multitask, freeze 3 epochs	multitask	97,332,362
arm03	Wav2Vec2 multitask, freeze 20 epochs	multitask	97,332,362
arm04	CNN + pooling, multitask	multitask	342,030
arm05	CNN + single head	single-head	274,950
arm06	CNN + LSTM, multitask	multitask	457,614
arm07	CNN + Transformer, multitask	multitask	540,302

4.2 Training Protocol

All models are trained on a single NVIDIA GPU with the following shared settings: seed $= 42$, 20 epochs with early stopping (patience $= 5$), mixed precision in the classification pipelines, batch size $= 8$ (binary classifiers), 16 (multitask and CNN classifiers), maximum audio length $= 3$ seconds, and sample rate $= 16$ kHz. No gradient clipping is applied; an earlier configuration that clipped the total gradient norm was abandoned because, on a 94.6M-parameter model with a randomly initialized head, the clip dominated the head's early gradient signal. Preprocessed audio is cached to disk for fast re-runs. We train each configuration once, from seed 42; every reported value is therefore a single-run point estimate.

Thresholds for binary classification are tuned on the held-out 20% subsample of the training pool described in Section 3 (the operating point is referred to as *validation-tuned* throughout) by sweeping a fixed grid of 17 decision thresholds (`THRESHOLD_SWEEP = np.arange(0.1, 0.91, 0.05)`: 0.10, 0.15, ..., 0.90) and selecting, per class, the value that maximizes F1 on that subsample—not Youden's J, which is computed and stored separately. For example, the multitask Wav2Vec2 model (arm02) uses thresholds of 0.55 (prolongation), 0.35 (block), 0.40 (sound repetition), 0.35 (word repetition), and 0.45 (interjection). No test-set or cross-corpus threshold fitting is performed. We note that F1-optimal operating points selected on this in-distribution subsample can still be degenerate under distribution shift: Section 5 reports large differences in predicted-positive rate when these thresholds are applied to the external Boli set. Because the subsample is drawn from the training pool rather than from a disjoint validation split, it is also the same data used for early-stopping model selection, so the tuned thresholds are mildly optimistic; we treat this as a limitation rather than a source of hidden test-set leakage, since the test split is never used for any selection.

4.3 Evaluation Metrics

For classification, we report precision, recall, F1 score, AUROC, and AUPRC per dysfluency type, plus macro-averaged F1 across all five types. Because the classes are imbalanced (the majority of clips is fluent), AUPRC complements AUROC as a threshold-independent measure of ranking quality on the positive (dysfluent) class. We report macro values as the unweighted arithmetic mean over the five classes.

4.4 Evaluation Protocol

Two evaluation settings are used, with distinct interpretations.

In-distribution, clip-level held-out (Test). The merged-corpus test split contains 3,716 clips: 3,662 clips from SEP-28K and UCLASS plus the 54 Project Boli clips described below. This is a *clip-level* split: for SEP-28K and UCLASS clips, speakers may appear in both training and test partitions, because the split operates on clips and not on speakers. As a result, these in-distribution numbers are not speaker-independent and are best viewed as an upper-bound benchmark under near-distribution conditions. The results reported in this paper cover the 3,715 clips for which every model produced a prediction (one clip was discarded during prediction processing; the Boli subset therefore contributes 53 scored clips, $\approx 1.4\%$ of the scored test set). Because these 53 Boli clips are out-of-domain but included in the scored test set, the "in-distribution" numbers contain a small, quantified out-of-domain component; this does not change the qualitative ordering reported below but should be kept in mind when comparing our macro values with single-corpus results.

Speaker-independent, cross-corpus held-out (Boli). 53 clips from the Project Boli dataset (one of the 54 pinned clips was discarded during prediction processing) are therefore entirely unseen during training. They were recorded by different speakers and under different recording conditions than SEP-28K and UCLASS, and they carry word-level rather than clip-level annotations. This provides a *speaker-independent* estimate of generalization to new speakers and recording conditions; it is not a cross-lingual evaluation, since all 54 Boli clips are English (Section 3). Per-class clip counts on Boli are limited (PR: 16, B: 35, SR: 34, WR: 14, IN: 9), so Boli results are noisy but honest.

5 Results and Discussion

5.1 In-Distribution and Cross-Corpus Macro F1

Table 3 summarizes macro F1 for all seven configurations under both default (0.5) and tuned thresholds, on the clip-level in-distribution test set and on the speaker-independent Boli set. Figures 1 and 2 show the corresponding per-class F1 scores for the multitask Wav2Vec2 model and the CNN models.

Under the clip-level in-distribution protocol, the *five-binary Wav2Vec2 classifier* (arm01) achieves the highest measured test macro F1 (0.5704 at tuned thresholds), and the *multitask Wav2Vec2* with 3-epoch backbone freezing (arm02) follows closely (0.5557) while using $\approx 4.9\times$ fewer parameters than five independent classifiers. The margin between the two top configurations is only ≈ 0.015. Because every configuration is a single training run, this difference is within the range of run-to-run variation we would expect from different random seeds, and we do not interpret it as evidence that one architecture is better than the other; both substantially outperform the remaining configurations under this protocol.

Table 3. Macro F1 across all seven configurations plus deterministic baselines. Test = clip-level in-distribution held-out; Boli = speaker-independent cross-corpus held-out. "0.5" uses a fixed 0.5 threshold; "tuned" uses per-class validation-F1-optimal thresholds (Section 4). "All-positive" always predicts the positive class (recall $= 1$ per class); "Random@rate" predicts positive with probability equal to the class base rate. Neither baseline requires tuning.

Arm	Test F1 (0.5)	Test F1 (tuned)	Boli F1 (0.5)	Boli F1 (tuned)
arm01_5x_w2v2	0.5730	0.5704	0.2540	0.2920
arm02_mt_frz3	0.5415	0.5557	0.1621	0.2384
arm03_mt_frz20	0.1413	0.3781	0.0457	0.2138
arm04_cnn_pool	0.1492	0.2556	0.3507	0.3315
arm05_cnn_single	0.2116	0.2617	0.4501	0.4756
arm06_cnn_lstm	0.2418	0.2489	0.4508	0.5077
arm07_cnn_tf	0.2694	0.2714	0.4214	0.3451
All-positive	0.247	0.247	0.550	0.550
Random@rate	0.143	0.143	0.408	0.408

Backbone-freezing duration matters: the *multitask Wav2Vec2 with 20-epoch freezing* (arm03) performs notably worse (test F1 $= 0.141$ at default, 0.378 at tuned), confirming that excessive backbone freezing prevents the model from learning task-specific representations.

Under the clip-level protocol, the CNN-based configurations (arms04–07) achieve substantially lower in-distribution macro F1 (0.149–0.269 at default thresholds), reflecting the limited capacity of small convolutional networks ($\sim$300K–540K parameters) relative to the 94M-parameter Wav2Vec2 backbone. For reference, an all-positive baseline and a base-rate-random classifier achieve macro F1 of 0.247 and 0.143 on the same test set: the Wav2Vec2 configurations clearly capture in-distribution signal, while the CNN configurations exceed these trivial baselines only marginally (tuned macro F1 0.25–0.27).

5.2 Speaker-Independent Evaluation (Project Boli)

On the 53-clip Boli set—whose speakers and recording conditions are fully unseen during training—a naive *all-positive* classifier, which predicts every clip as positive for every dysfluency type, achieves a macro F1 of 0.550, *higher than every configuration we evaluated* (best: arm06, 0.5077). Table 4 reports per-class tuned F1 for all seven configurations together with the all-positive baseline. Read naively, the table appears to show that the CNN configurations "generalize best" to the unseen corpus; however, none of them exceeds the trivial baseline, and the per-class picture shows why. The CNN configurations reach comparatively high F1 on block, word repetition, and interjection almost entirely by predicting (nearly) every clip as positive: at its validation-tuned thresholds, arm06 applies the positive label to essentially all 53 Boli clips for block, word repetition, and interjection, so its per-class F1 approaches—and for interjection exactly equals—the all-positive F1 that is achievable without any learning (Table 5). By contrast, the Wav2Vec2 configurations collapse toward *all-negative* predictions on the shifted distribution: their macro predicted-positive rate (PPR) is 0.08–0.10 against an

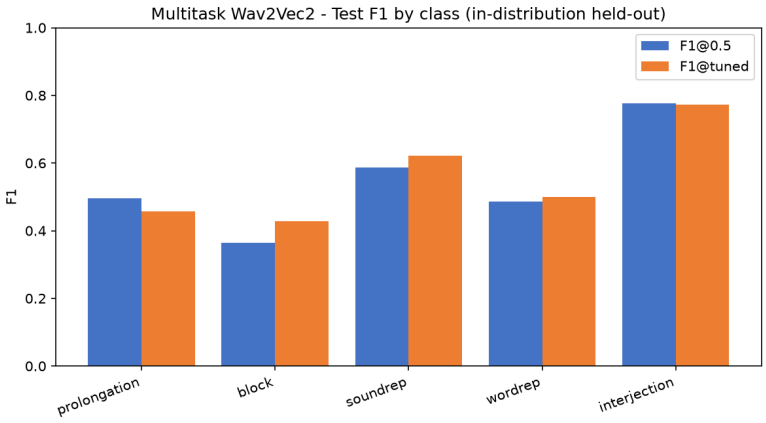


Fig. 1. Per-class F1 scores (multitask Wav2Vec2) on the clip-level in-distribution test set.

Table 4. Per-class tuned F1 on the Project Boli speaker-independent held-out set (PR = prolongation, BL = block, SR = sound repetition, WR = word repetition, IN = interjection). Class support on Boli is small (PR: 16, BL: 35, SR: 34, WR: 14, IN: 9). "All-positive" labels every clip as positive for every class.

Arm	PR	BL	SR	WR	IN	Macro
arm01_5x_w2v2	0.400	0.250	0.256	0.125	0.429	0.292
arm02_mt_frz3	0.200	0.154	0.256	0.182	0.400	0.238
arm03_mt_frz20	0.200	0.233	0.182	0.182	0.273	0.214
arm04_cnn_pool	0.308	0.491	0.182	0.387	0.290	0.332
arm05_cnn_single	0.326	0.765	0.508	0.431	0.348	0.476
arm06_cnn_lstm	0.492	0.795	0.542	0.418	0.290	0.508
arm07_cnn_tf	0.316	0.320	0.490	0.267	0.333	0.345
All-positive	0.464	0.795	0.782	0.418	0.290	0.550

average class base rate of 0.41, so their F1 is limited by near-zero recall on most classes. The tuned-F1 "reversal" reported in the in-distribution section (Table 3) is therefore an artifact of operating-point drift under distribution shift, not evidence of a difference in generalization ability.

The threshold-free metrics corroborate this reading. Table 5 reports macro AUROC and macro AUPRC for all configurations. The two task-adapted Wav2Vec2 configurations (arm01, arm02) rank Boli clips best: macro AUROC 0.620–0.621, and macro AUPRC 0.546–0.557, i.e., a lift of ≈ 1.3–$1.4\times$ over the base-rate prior. The frozen-backbone variant (arm03) sits near chance (AUROC 0.538, lift 1.09). The CNN configurations rank around chance: macro AUROC 0.473–0.522 and macro AUPRC 0.412–0.445 (lift ≈ 1.01–1.09). In absolute terms the Wav2Vec2 ranking advantage is modest (e.g., macro AUROC 0.621 vs. 0.522; each class has at most 35 positive clips, nine in the worst case), so no strong generalization claim is warranted in either direction; but the direction of the evidence is the *opposite* of the F1-based reading: under threshold-

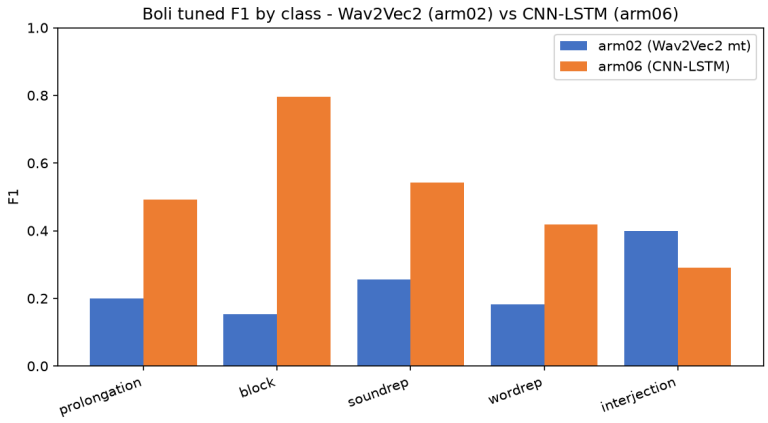


Fig. 2. Per-class tuned F1 on the Project Boli speaker-independent cross-corpus set for the shared-backbone Wav2Vec2 model (arm02) and the CNN-LSTM model (arm06), each using its per-class validation-F1-optimal thresholds.

free metrics the Wav2Vec2 models separate dysfluent from fluent clips on the unseen corpus better than the CNN models do, and none of the seven configurations comes close to reliable cross-corpus detection.

The primary open question is therefore not "which representation family generalizes better"—the present data do not support a CNN advantage once base rates and operating points are accounted for—but *why* validation-calibrated operating points drift so strongly under distribution shift, and whether the modest ranking advantage of Wav2Vec2 features persists on larger external corpora. We frame two candidate explanations, both of which the present experiments support only circumstantially and do not directly test:

- *Operating-point drift hypothesis:* the shifted Boli distribution changes the models' score distributions (score matching on out-of-domain inputs), so fixed validation-tuned thresholds land in degenerate regions of the ROC curve. Because F1 depends on the operating point, it then tracks base rates and the predicted-positive rate rather than discrimination; threshold-free metrics are immune to this effect.
- *Corpus-specificity / spectrogram-invariance hypothesis (ranking):* the higher-capacity Wav2Vec2 representations may encode corpus-specific characteristics of SEP-28K/UCLASS—speakers, accents, recording conditions—that transfer only weakly to unseen speakers, while the coarse spectral patterns preserved by mel-spectrogram CNNs transfer more evenly; this would explain why the two families differ in ranking quality the observed direction without resolving which family is preferable for a given deployment.

Both hypotheses can be tested experimentally without retraining the classifiers: lightweight *probe* classifiers trained on the frozen embeddings of each model family—one predicting speaker identity and one predicting the source corpus—would test the

Table 5. Threshold-free macro metrics and predicted-positive rate (PPR) on the Project Boli set (53 clips). AUPRC of a random classifier equals the average base rate (macro 0.408), so the "lift" column reports AUPRC relative to that prior. PPR is the mean over classes of the fraction of clips predicted positive at the validation-tuned thresholds.

Arm	Macro AUROC	Macro AUPRC	AUPRC lift	Macro PPR
arm01_5x_w2v2	0.620	0.546	1.34	0.079
arm02_mt_frz3	0.621	0.557	1.37	0.102
arm03_mt_frz20	0.538	0.443	1.09	0.204
arm04_cnn_pool	0.473	0.412	1.01	0.574
arm05_cnn_single	0.497	0.445	1.09	0.615
arm06_cnn_lstm	0.522	0.419	1.03	0.879
arm07_cnn_tf	0.500	0.412	1.01	0.438
Base rate (random)	0.500	0.408	1.00	0.408

Table 6. Per-class metrics for the multitask Wav2Vec2 (arm02) on the clip-level test set at tuned thresholds. Macro is the unweighted mean over classes.

Class	F1 (tuned)	AUROC	AUPRC	Support
Prolongation	0.457	0.845	0.491	400
Block	0.429	0.768	0.444	553
Sound Rep.	0.621	0.878	0.624	494
Word Rep.	0.499	0.855	0.550	366
Interjection	0.773	0.935	0.848	834
Macro	**0.556**	**0.856**	**0.592**	—

corpus-specificity and spectrogram-invariance hypotheses, while recalibrating thresholds on a handful of Boli clips would directly quantify operating-point drift. We leave these probe experiments to future work. The Boli evaluation set is small (53 clips; per-class support as low as 9), so definitive conclusions about cross-corpus generalization require additional external corpora.

5.3 Per-Class Analysis

Detailed per-class metrics for the multitask Wav2Vec2 (arm02) on the clip-level test set at tuned thresholds are presented in Table 6, including AUPRC. The tuned thresholds themselves are listed in Table 7; for the multitask model these are per-class validation-F1-optimal operating points (prolongation 0.55, block 0.35, sound repetition 0.40, word repetition 0.35, interjection 0.45). Figure 3 shows the per-class AUROC and AUPRC.

Several observations follow.

- *Interjection* achieves the highest F1 among the evaluated classes (F1 $= 0.773$, AUROC $= 0.935$, AUPRC $= 0.848$). Filler words ("um," "uh," "like") have distinct acoustic signatures that are well-separated from fluent speech, making them relatively easy for the model to identify.
- *Block* is the most challenging class (F1 $= 0.429$, AUROC $= 0.768$, AUPRC $= 0.444$). Silent pauses and tense pauses before speech are acoustically similar

Table 7. Per-class validation-tuned thresholds (F1-optimal over the swept grid) for the multitask Wav2Vec2 (arm02).

Class	Threshold
Prolongation	0.55
Block	0.35
Sound Rep.	0.40
Word Rep.	0.35
Interjection	0.45

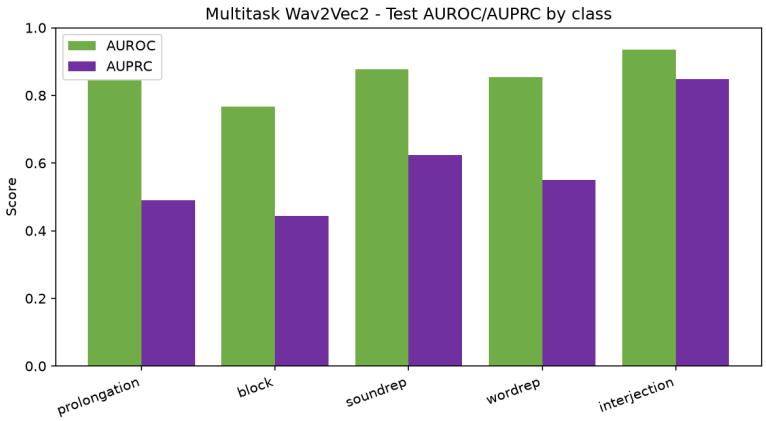


Fig. 3. Per-class AUROC and AUPRC (multitask Wav2Vec2) on the clip-level test set.

to normal speech pauses, making them difficult to distinguish without contextual information. The fixed 3-s window and silence trimming described in Section 3 may additionally suppress the very pause that defines a block when it falls near a window boundary.

– *Sound repetition* achieves the second-highest F1 (0.621) with the second-highest recall (0.627), indicating the model is effective at catching sound repetitions but with lower precision.
– The gap between AUROC and F1 (e.g., prolongation AUROC $= 0.845$ vs. F1 $= 0.457$) indicates that the model ranks dysfluent clips well but its probability outputs are not well matched to the validation-tuned operating thresholds when those thresholds are applied to the test set. We do not measure calibration directly; quantifying predictive calibration (e.g., expected calibration error and Brier score, following Guo et al. [8]) is an explicit item of future work.

Regarding blocks specifically, a natural hypothesis is that explicit temporal and prosodic features—pause duration, voicing, and energy trajectory around the suspected pause—would improve block detection beyond what the raw classifiers extract. We regard this as the most promising single-class improvement and propose it as a dedicated follow-up study; the present work only isolates the failure pattern.

Table 8. Confusion matrices (TP, FP, TN, FN) for the five binary Wav2Vec2 classifiers (arm01) on the clip-level test set at tuned thresholds. Each row sums to the 3,715 scored clips.

Class	TP	FP	TN	FN	Threshold
Prolongation	189	139	3176	211	0.550
Block	210	284	2878	343	0.500
Sound Rep.	301	185	3036	193	0.450
Word Rep.	182	151	3198	184	0.400
Interjection	668	175	2706	166	0.500

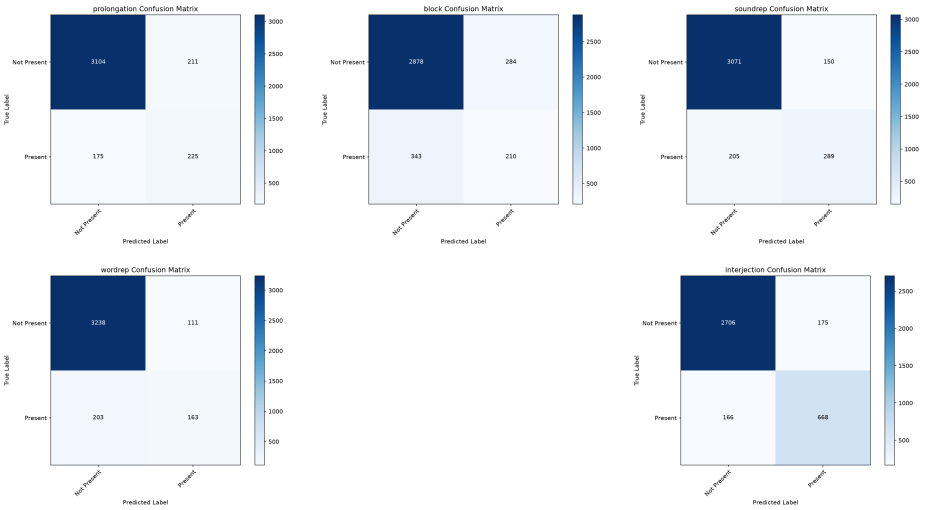


Fig. 4. Confusion matrices for the five binary Wav2Vec2 classifiers (arm01) on the clip-level in-distribution test set. Top row: prolongation, block, and sound repetition; bottom row: word repetition and interjection.

Table 8 reports confusion matrices (TP, FP, TN, FN at tuned thresholds) for the five binary Wav2Vec2 classifiers (arm01) on the test set, and Figure 4 shows them graphically. They show the same pattern as the multitask model: interjection is detected most confidently, block produces the most false negatives and the largest number of false positives relative to its support, and prolongation is the next most frequently missed.

5.4 Computational Considerations

Table 9 summarizes the parameter-accuracy-generalization trade-off for the most informative configurations. Sizes are computed from the reported parameter counts assuming 32-bit floating-point storage (MB $=$ params $\times\, 4/10^6$); we did not benchmark GPU memory, latency, or throughput, and report only the analytically derived footprint.

Wav2Vec2 models achieve $\sim$2$\times$ higher in-distribution F1 than the CNN models but comprise roughly 180$\times$ (multitask) to 875$\times$ (five binary classifiers) more parameters. On the external Boli set their tuned F1 is lower, but Section 5 shows that this reflects

Table 9. Trade-off between parameter count, computed fp32 size, and evaluation performance (tuned F1, Table 3). On Boli, an all-positive baseline reaches macro F1 0.550, above every configuration shown here (Section 5).

Model family	Params	Size (fp32)	Test F1	Boli F1
Wav2Vec2 (5× binary)	472.8M	1.89 GB	0.570	0.292
Wav2Vec2 (multitask)	97.3M	389 MB	0.556	0.238
CNN (best)	540K	2.2 MB	0.271	0.508

operating-point drift and class-base-rate effects rather than a ranking advantage: under threshold-free metrics the Wav2Vec2 configurations rank Boli clips *better* than the CNNs (macro AUPRC 0.546–0.557 vs. 0.412–0.445). From an engineering standpoint, the multitask Wav2Vec2 configuration delivers nearly the same clip-level F1 as five independent classifiers at a fraction of the parameter budget. The CNN models remain attractive for resource-constrained settings given their very small footprint, but they are not a drop-in generalization remedy: none of the evaluated configurations exceeds a trivial baseline on the external corpus, so cross-corpus deployment of any of these systems is not yet supported by evidence.

5.5 Comparison with Published Results

Our best Wav2Vec2 results (macro F1 $= 0.570$ at tuned thresholds for the five-binary configuration, 0.556 for the multitask variant, clip-level split) sit within the range reported in the literature: Bayerl et al. [4] reported macro F1 of 0.56–0.63 on FluencyBank, and Sušac et al. [15] reported per-class F1 from 0.30 (block) to 0.78 (interjection) on SEP-28K with an attention-based multiple-instance-learning model. Our per-class values follow the same ordering, with interjection highest (0.773 for the multitask variant) and block lowest (0.429); our measured orderings are consistent with that of Sušac et al.—for example, our block F1 is lower than our interjection F1, and both are in the same relative arrangement. However, differences in datasets, splits, label definitions, threshold strategies, and evaluation protocols prevent direct attribution of absolute differences to architecture alone, and we make no such claim.

5.6 Limitations

- *Clip-level in-distribution split:* The test set shares speakers with training for SEP-28K and UCLASS, inflating in-distribution metrics; those results are not speaker-independent. The only speaker-independent evidence is the Boli cross-corpus set, which is small (53 clips).
- *Single seed:* All results are single training runs from seed 42. Statistics such as mean $\pm$ standard deviation or confidence intervals require multi-seed runs and are not available; accordingly we do not claim statistically significant differences, especially between arm01 and arm02, whose measured difference (0.015) is small.
- *Loss differs across representation families:* the Wav2Vec2 configurations are trained with focal loss ($\gamma = 2.0$), whereas the CNN configurations are trained

with positive-weighted binary cross-entropy (Section 3). Family-level comparisons therefore confound representation with training objective; isolating the two would require training each family under both objectives.
- *Threshold tuning reuses the training pool:* the prepared 3,662-clip validation split is unused; early-stopping model selection and threshold tuning both operate on the same 20% stratified subsample of the training clips (5,857 clips, seed 42). The tuned operating points are therefore mildly optimistic, and the two selection steps are not statistically independent. The test split is never used for any selection, so this is a limitation of the tuning protocol rather than test-set leakage.
- *English-only cross-corpus set:* all 54 ingested Boli clips are English, so our cross-corpus protocol tests speaker, recording-pipeline, and annotation-convention shift but not language shift. Cross-lingual robustness remains untested and should not be inferred from these results.
- *Small external corpus:* Boli provides 53 scored clips, with per-class supports of at most 35; the cross-corpus conclusions are therefore noisy.
- *Operating-point dependence:* All tuned-F1 results depend on validation-F1-optimal thresholds. On Boli these thresholds drift strongly (PPR ranges from 0.08 for the Wav2Vec2 configurations to 0.88 for the CNN configurations), and no configuration exceeds an all-positive baseline (macro F1 0.550) on that set. Cross-corpus *deployment* of any evaluated system is therefore not supported; threshold-free ranking is reported alongside tuned F1 for this reason.
- *Untested hypotheses:* The corpus-specificity, spectrogram-invariance, and operating-point-drift explanations are not directly tested; embedding-based probe experiments (speaker-ID and source-corpus classifiers) and threshold recalibration probes are proposed as future work.
- *Scope of baselines:* Only Wav2Vec2-based and compact CNN models are evaluated. Additional self-supervised backbones (HuBERT, XLS-R, WavLM), a classical acoustic baseline (e.g., MFCC + SVM), and further external corpora would strengthen the comparison and are left to future work.
- *Calibration:* Predictive calibration (e.g., ECE, Brier score) is not measured; per-class thresholds are tuned on validation only and may not transfer to all deployment scenarios.

5.7 Practical Implications

The results indicate that model selection cannot be grounded in in-distribution metrics alone: the threshold-dependent F1 ranking reverses on the unseen corpus through operating-point drift rather than through a real ranking advantage, and no configuration exceeds a trivial baseline on the external set. Until speaker-independent evidence is obtained with additional corpora and multiple seeds, deployment guidance should remain conservative:

- *Report both protocols and both metric families always:* clip-level in-distribution and cross-corpus metrics should be reported together, and threshold-dependent metrics (F1) should be accompanied by threshold-free metrics (AUROC, AUPRC) and trivial baselines, since any one of these alone under- or overstates performance.

- *Inflated-deployment caution:* the in-distribution test shares speakers with training; any system deployed on new speakers will likely perform below the reported in-distribution numbers.
- *Interjection* achieves the highest F1 among the evaluated classes and can serve as a cheap diagnostic signal, but it should not be equated with reliability of the remaining classes; block detection in particular requires complementary signals (e.g., pause-aware features).
- *Document the split type:* whether speakers overlap between training and test materially changes how results may be interpreted.

6 Conclusion

This paper presented an end-to-end speech dysfluency classification system and compared seven experimental configurations across two representation families under a clip-level in-distribution protocol and a speaker-independent cross-corpus protocol. Our main observations are:

- Under the clip-level in-distribution protocol, the *five-binary Wav2Vec2 classifier* achieved the highest measured macro F1 (0.5704), while the *shared-backbone multitask variant* followed closely (0.5557) using $\approx 4.9\times$ fewer parameters in total; the difference between the two is small and should be treated as a single-seed point estimate.
- On the Project Boli speaker-independent cross-corpus set, no configuration exceeds an all-positive baseline (macro F1 0.550). The apparent F1 advantage of the CNN configurations is an operating-point artifact—their validation-calibrated thresholds drift toward near-all-positive predictions (macro PPR 0.88) while the Wav2Vec2 thresholds drift toward near-all-negative (macro PPR 0.08). Under threshold-free metrics the ranking reverses and becomes modest: the Wav2Vec2 configurations reach macro AUPRC 0.546–0.557 (≈ 1.3–$1.4\times$ the base-rate prior) versus 0.412–0.445 for the CNN configurations.
- *Interjection* achieved the highest F1 among the evaluated classes (0.773), while *block* was the most challenging (0.429), consistent with the ordering reported elsewhere in the literature.
- *Backbone freezing duration* materially affects the multitask Wav2Vec2 configuration: freezing for 20 epochs instead of 3 degrades tuned macro F1 by $\approx 32\%$.

The tension between the two protocols—Wav2Vec2 dominating in-distribution and ranking slightly better on the unseen corpus under threshold-free metrics, yet suffering far larger operating-point drift—motivates further investigation into the relationship between representation capacity, corpus-specific information encoded by a representation, and cross-corpus robustness. We specifically propose: (1) speaker-independent splits of the full corpus with multi-seed runs and confidence intervals; (2) embedding-based probe experiments (speaker-identification and source-corpus classification) together with threshold recalibration probes to test the corpus-specificity, spectrogram-invariance, and operating-point-drift hypotheses; (3) additional external corpora and languages; (4) evaluation of HuBERT, XLS-R, and WavLM backbones as well as a

classical MFCC baseline; (5) quantitative calibration analysis and pause-aware features for block detection.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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